

Speech-To-Text and NLP: Comparing De Gasperi and Meloni's Rhetoric

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This report compares the rhetoric of Alcide De Gasperi and Giorgia Meloni. After building a corpus of public speeches transcribed with Whisper from Giorgia Meloni's videos on YouTube, I compare them with a corpus of speeches from the Italian Republic's first Prime Minister Alcide De Gasperi. The results show that De Gasperi's speeches are, on average, 30% more diversified in their lexicon, and the corpora have a Jaccard similarity index of 32%. The highest-frequency words in common show transversal themes, such as freedom, peace and nation, while the highest-frequency words not in common reflect the PMs' different political and societal landscapes. I also performed a sentiment analysis on each speech using a BERT classifier. The results underscore the importance of domain-appropriate training data: the model was trained on product reviews, not on political rhetoric, which translated to an average confidence of roughly 40% per classification.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Speech recognition**; • **Social and professional topics** → **Political speech**; • **Information systems** → **Sentiment analysis**.

Additional Key Words and Phrases: AI, NLP, Politics, Speeches

1 INTRODUCTION

Politicians are some of the most influential people in human society: from the Greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page (1984) [13] showed how nearly all 20th century U.S. Presidents have tried to lead public opinion, with varying degrees of success. As societies became more democratic, so grew the leaders' responsibility to convince potential followers that their policies are beneficial. Thus, speeches play a vital role in the functioning of society [2]: leaders depend on the verbal power to persuade people [4].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et al. [6] evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump attacking the legitimacy of the 2020 US presidential election. They found no evidence indicating that elites can erode democratic norms easily or that the effects of norm violations are uniform across the entire population. However, elite rhetoric can shape normative beliefs in core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office.

Having established the importance of speeches, we can differentiate between two types: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social network posts. This last category is especially interesting since it affords politicians a direct and high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted 57,000 times between May 2009 and January 2021 alone [10], while Italy's Prime Minister Giorgia Meloni totalled more than eighty million social network interactions in 2025 [17].

On the other hand, internal speeches are especially easy to access, being already aggregated in public databases, yet their technical and codified nature makes their interpretation harder for both human and automated agents. Moreover, until 1985 Italian stenographic documentation did not report speeches word for word, but rather translated them into a "language of the parliament" [8, 12].

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2 SOCIAL NETWORKS

Broadly speaking, the goal of this mini-project is to gather a corpus of politicians' speeches and perform linguistic analyses on it. The idea is to start from standard NLP tasks such as word frequency and text complexity, to then move to classification tasks such as sentiment analysis and hate speech detection.

This objective is why I ultimately decided to gather a corpus of *external speeches* rather than internal ones. Large language models are trained with tokenizers, and the resulting token distribution is highly imbalanced: Chung et al. [5] did a controlled study that scaled the vocabulary of the language model from 24K to 196K while holding data, computation, and optimization unchanged. They discovered that models are disproportionately optimized on the part of language that appears most often, meaning they are way better at understanding common words and patterns. This translates to the fact that internal speeches are less understandable for an AI agent.

As previously stated, external speeches include social network communications. While television is still the first source of information for the majority of the population, online platforms are steadily growing and are already the preferred media for young people [7, 11]. There is evidence of an increase in political participation due to social media usage, as well as risks for the functioning of democracy [1, 9], and there have been attempts to predict election results with social media data analyses. Rita et al. measure sentiment polarity on Twitter, and conclude that tweets' sentiment is not a reliable election results predictor. Additionally, results also show that it is impossible to state that social media impacts voting decisions [14].

Opposite results can be seen in Belcastro et al.'s [3] 2022 study: a real-time analysis was carried out during the 2020 US presidential election campaign, correctly identifying the leading candidate before Election Day in 10 out of 11 swing states.

Silva et al. [15] managed to match tweets against parliamentary speeches to measure politicians' sentiment on a specific topic. They find that parliament members who participate less in parliamentary debate tend to have larger differences with their party on Twitter, suggesting that a certain level of self-censoring is taking place in the parliamentary arena.

That being said, X is just one channel among many used to connect with the voters: official messages to the nation, interviews, and public speeches are still powerful tools, and are often made publicly available in video format by the politicians themselves on their Facebook page and YouTube channel.

3 PROJECT GOAL

Let's now define the goal of this project.

I want to compare the rhetoric between the first and the current¹ Prime Ministers of the Italian Republic, namely Alcide De Gasperi and Giorgia Meloni. The rhetoric will be compared using natural language processing techniques, such as measuring textual complexity and sentiment analysis. Only public speeches will be taken, thus leaving out all parliamentary speeches.

I also want to produce a proof-of-concept for a speech-to-text pipeline that, given a set of URLs, each of which is a YouTube's permalink, transcribes the video into text.

The first step will be gathering two corpora of public speeches, one for each PM. In the case of Giorgia Meloni, I want to compile the corpus by transcribing video interviews and public talks available on her official YouTube channel.

¹At the time of writing: May 2026.

Then, after having gathered the corpora into suitable machine-readable data (such as csv), I will pre-process them with procedures such as tokenization, stemming and lemmatization. Lastly, I will compute word frequency and text complexity measures, as well as a sentiment analysis.

4 BUILDING THE CORPUS

4.1 Alcide De Gasperi's Corpus

I would like to thank Sara Tonelli for providing me with the De Gasperi's Corpus [16], a collection of Alcide De Gasperi's public documents with gold and silver annotation.

The corpus is a collection of 2,762 documents issued between 1901 and 1954, formatted into XML files which include metadata that covers not only the title, the date and the place of publication, but also key-concepts automatically extracted from each text (with the corresponding relevance score) and genre labels manually assigned by domain experts. Furthermore, the release includes silver annotation for lemma, part of speech, person names and place names with associated coordinates in a CoNLL-like format.

An example of the XML formatting can be seen at listing 1. Thanks to the `<genres>` tag, I was able to extract 474 public speeches, each of which with a location, a precise date and a list of keywords.

Listing 1. An example of how a single speech in De Gasperi's Corpus is formatted in XML

```
1 <?xml version="1.0" encoding="UTF-8" standalone="no"?>
2 <document id="I.doc_7">
3   <publication_date>1901-11-26</publication_date>
4   <publication_place>
5     <place_name>Trento</place_name>
6     <place_latitude>46.0664228</place_latitude>
7     <place_longitude>11.1257601</place_longitude>
8   </publication_place>
9   <keywords>
10    <keyword score="39.16">first keyword</keyword>
11    ...
12    <keyword score="5.59">last keyword</keyword>
13  </keywords>
14  <genres>
15    <genre>Speech Type</genre>
16  </genres>
17  <text>The speech itself is stored between these brackets</text>
18 </document>
```

4.2 Giorgia Meloni's Corpus

My objective was to build Meloni's corpus by transcribing videos of her public speeches, such as talks and interviews. Fortunately, there is an unofficial YouTube channel (that I reached from the Prime Minister's official website) which aggregates more than four thousand videos of her public appearances. The channel is called "Giorgia Meloni News"².

Given a YouTube URL, I can use Python's library *yt-dlp* to retrieve the video's metadata and download its audio content in mp3 format, as shown in listing 2. I had to manually pass the cookies taken from my web-browser, and I used some extra commands to prevent YouTube to block the requests due to suspicious activity.

²Giorgia Meloni News: <https://www.youtube.com/@GiorgiaMeloniTv>

Then, the mp3 file just retrieved gets injected into OpenAI's Whisper library, which uses an encoder-decoder Transformer to transcribe it into text, as shown in listing 3. The model I used is the "medium"³, which at 5 GB fits within the total 6 GB of VRAM available to my RTX 4050 GPU. The model was instructed to predict Italian.

The difference in precision between the models "tiny" (which requires less than 1 GB of VRAM) and "medium" is quite high, as can be seen in the following pieces of text that have been generated from the same audio file:

- **Model tiny:** "iamo con se è beruto fuori da questa rio neone sotto il profiro tecnico per quanto riguarda la franha e come il governo un tino era ad essere vicino alla popolazione di Nishenia."
- **Model medium:** "Diciamo cosa è venuto fuori da questa riunione sotto il profilo tecnico per quanto riguarda la frana e come il governo continuerà ad essere vicino alla popolazione diniscemeca."

The last step of the pipeline saves the transcription into a csv file, along with the extracted metadata. Each file has the following fields: *politician* ("meloni" in this case), *historical_date* (the upload date of the video), *location* and *tags* (extracted from the metadata if available, empty strings otherwise), *description* and *title* of the video, *url* which stores the permalink of the video itself, and lastly *text* which holds the whole transcription.

Each video is processed immediately after being retrieved, and until it has been saved in a csv file the program does not try to fetch the next one. This is done for two reasons: first, to prevent *yt-dlp* from sending requests too close to each other, which could result in YouTube denying them due to suspicious activity. Second, even if the pipeline halts for any reason, the videos processed up to that point will not be lost.

Listing 2. Downloading audio and metadata from a YouTube video

```
1 result = subprocess.run([
2     "yt-dlp",
3     "--print-json",
4     "-x",                                # download audio only
5     "--audio-format", "mp3",
6     "--cookies", "yt_cookies.txt",      # manual cookies, extract with browser extension
7     "--sleep-requests", "2",           # Sleep 2s between requests
8     "--sleep-interval", "5",           # Sleep 5s between downloads
9     "--max-sleep-interval", "15",       # Randomize up to 15s
10    "--limit-rate", "5M",               # Throttle to 5MB/s (mimics streaming)
11    "-o", audio_file.replace('.mp3', ''),
12    video_url
13 ], capture_output=True, text=True, check=True)
```

Listing 3. Transcribe the audio file into text using Whisper

```
1 subprocess.run([
2     "whisper",
3     "--language", lang_code,           # "Italian" chosen
4     "--word_timestamps", "True",
5     "--model", model_name,             # model "medium" chosen
6     "--output_dir", output_dir,
7     "--device", "cuda",               # ensures the model gets loaded in GPU
8     audio_file
9 ], check=True)
```

³Whisper model sizes available are tiny, base, medium and large.

4.3 Corpora Description

Table 1 describes the corpora collected so far: the date each corpus was compiled, the number of speeches contained, the period in which the speeches were originally delivered (represented as a date range, from the oldest to the most recent speech), and the median speech length in words

Table 1. Summary of Meloni and De Gasperi corpora.

Metric	Meloni Corpus	De Gasperi Corpus
Speeches Date Range	2025-05-16 → 2026-05-21	1901-09-21 → 1954-07-24
Number of Speeches	72	473
Median Words/Speech	982	701
Compiled On	May 2026	July 2019

5 TEXTUAL ANALYSIS

All the code used to compute the results shown in the next sections can be seen in the Jupyter notebook [mhetacc/cross-dem/notebooks/text_analysis.ipynb](https://github.com/mhetacc/cross-dem/notebooks/text_analysis.ipynb).

5.1 Token Count and Textual Complexity

The collected corpus contains 473 speeches from De Gasperi and 72 from Meloni, with a total token count of 609,142 for De Gasperi and 120,116 for Meloni. On average, Meloni’s corpus has 1668 tokens per speech, while De Gasperi’s has 1287 tokens per speech. Word metrics are shown in table 2. The text was tokenized with spaCy’s Italian model “it_core_news_sm”.

I then measured the textual lexical diversity using *MTLD*, which I preferred over the more common *TTR* (Type-Token Ratio) since *MTLD* is agnostic to corpus size. I used the *LexicalRichness* library⁴ to compute the *MTLD* for each speech, and then both the average and the median *MTLD* for both PMs. Results can be seen in table 3.

To better visualize the difference in textual diversity, I plotted the *MTLD* values of 70 different speeches of Meloni and De Gasperi against each other. The plot can be seen in figure 1.

Table 2. Total tokens and average tokens per speech for De Gasperi and Meloni.

Politician	Total Tokens	Speeches Count	Avg Tokens/Speech
De Gasperi	609,142	473	1,287.83
Meloni	120,116	72	1,668.28

Table 3. *MTLD* metrics for De Gasperi and Meloni.

Politician	Avg <i>MTLD</i>	Median <i>MTLD</i>
De Gasperi	141.830	137.642
Meloni	106.745	104.918

⁴LexicalRichness is a small Python module to compute textual lexical richness measures: <https://pypi.org/project/lexicalrichness/>.

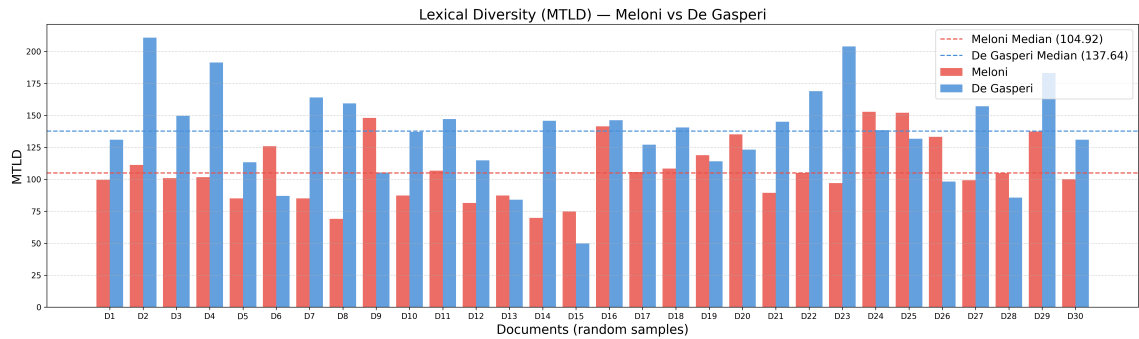


Fig. 1. MTLD values for 60 different speeches (30 for each Prime Minister), randomly sampled. On the Y-axis the MTLD values are reported, while on the X-axis there are the relative IDs for the pairs of speeches. Each pair of speeches is represented by two vertical bars (one red for Meloni, one blue for De Gasperi). Bar height represents their respective speech's MTLD value. The medians for both Prime Ministers are shown by the two horizontal dotted lines.

5.2 Vocabulary Comparison

I then wanted to see differences and similarities in the two Prime Ministers' vocabulary. I thus extracted from the corpus a filtered subset of tokens (example in listing 4), then I used the *Counter* method from the *collections* library to count the 1000 most commonly used words for both PMs.

Of the top-1000 words, 487 are in common, with a Jaccard similarity of 32.19%.

After extracting the top-1000 words by usage frequency for each PM, I cross-compared them. The results are shown in table 4. The first column shows words that are in common, whose compound frequency (*meloni_freq[word] + degasperi_freq[word]*) is highest. The second column shows words that are only found in De Gasperi's corpus (Meloni's frequency for those words is zero). The third column shows words that are only found in Meloni's corpus (De Gasperi's frequency for those words is zero).

It is interesting to see that words in common are about transversal themes, such as *italia*, *popolo*, *libertà*, and *democrazia*. Interestingly, *europa* is a top-20 shared word, showing us that the theme was and still is relevant even though the two PMs were born almost 100 years apart from each other.

On the other hand, the PM-specific words show us how the public discourse changes in the different periods: there are some themes that seem to have disappeared; for example, in the De Gasperi-only top 20 we see words such as *comunismo*, *socialisti*, and *marshall*, all of which were extremely relevant 70 years ago. In the Meloni-only top 20 we find words that simply could not have mattered back then, even though today they are absolutely central, such as *trump*, *ucraina*, *putin*, and *hamas*.

We can also see some examples of how the language itself changes over time: De Gasperi used quite often the word *dinanzi*, while Meloni often uses *dopodiché*. Both of these words are never used by the other Prime Minister.

To try to visualize how close (or far apart) the two PM's vocabularies are, I visualized their top-1000 words in a t-SNE scatter plot using the *scikit-learn* library. The results (in figure 2) show us that, with the exception of the words in common (in purple), the two vocabularies seem to be quite far apart, clustering in different regions.

Listing 4. Extract and filter words with tokenize function.

```
1 def tokenize(text):
2     return [token.text.lower() for token in nlp(text)]
```

```

3 if token.is_alpha      # only alphanumeric
4 and not token.is_stop  # no stop-words
5 and len(token)>3       # removes articles and some filler words
6 ]

```

Table 4. Comparison of most frequent words common to both PM. Last two columns show words specific to only one PM.

Rank	Common Words (Sum)		De Gasperi Only (Freq)		Meloni Only (Freq)	
1	italia	(1967)	gasperi	(458)	euro	(88)
2	popolo	(1201)	comunisti	(412)	ucraina	(85)
3	libertà	(1071)	trieste	(309)	sottotitoli	(65)
4	guerra	(927)	comunista	(241)	dopodiché	(48)
5	politica	(912)	avvenire	(237)	trump	(46)
6	pace	(910)	togliatti	(207)	infrastrutture	(37)
7	italiano	(828)	trento	(173)	approccio	(32)
8	presidente	(733)	comunismo	(171)	competitività	(30)
9	democrazia	(708)	socialisti	(170)	gaza	(28)
10	partito	(682)	oratore	(165)	migrazione	(28)
11	bisogna	(661)	deputati	(158)	scenario	(26)
12	italiani	(605)	trentino	(149)	centrodestra	(26)
13	soprattutto	(567)	dittatura	(145)	record	(23)
14	europa	(563)	democratici	(143)	hamas	(23)
15	sociale	(548)	nenni	(142)	palestina	(20)
16	cioè	(535)	voti	(137)	meloni	(19)
17	altra	(530)	degasperi	(127)	food	(19)
18	amici	(522)	onorevole	(126)	mattoni	(19)
19	nazione	(508)	dinanzi	(125)	buongiorno	(19)
20	senso	(502)	marshall	(122)	putin	(19)

5.3 Limitations

A better filtering of the extracted words would be needed: for example, "gasperi" and "degasperi" should ideally be merged into the bigram "de gasperi", if not removed entirely, since he did not refer to himself in the third person.

This could be achieved by applying named entity recognition (NER) to identify and merge multi-word names, or by post-processing the token list to combine frequent bigrams corresponding to named entities.

Additionally, a custom stop-word list could be compiled: for example, in Meloni's corpus the word "sottotitoli" appears often, but was likely added in post-production by the video uploader.

6 SENTIMENT ANALYSIS

The last analysis I performed on the dataset was a sentiment analysis using the *transformers* library from Hugging Face.

Originally I wanted to use *MilaNLProc/feel-it-italian-emotion*⁵, a model trained on the FEEL-IT corpus, made of Italian Twitter posts annotated with four basic emotions: anger, fear, joy, sadness. Unfortunately, at the time of writing the model's tokenizer is not compatible with any Python version I tried.

⁵<https://huggingface.co/MilaNLProc/feel-it-italian-emotion>

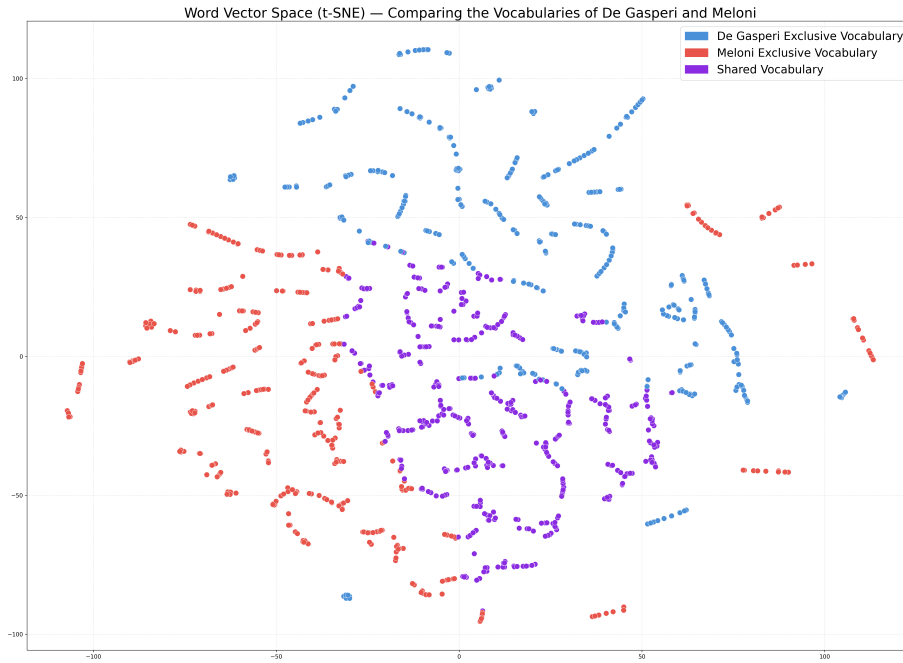


Fig. 2. t-SNE projection of the top-1000 most frequent words for each Prime Minister, converted into vector embeddings and plotted in a two-dimensional space. Red points represent words exclusive to Meloni, blue points words exclusive to De Gasperi, and purple points words shared by both

I opted for one of the most popular free models, *nlptown/bert-base-multilingual-uncased-sentiment*⁶, a bert-base-multilingual-uncased model fine-tuned for sentiment analysis on product reviews in six languages: English, Dutch, German, French, Spanish, and Italian. It predicts the sentiment as a number of stars (between 1 and 5).

I used a *pipeline* from the *transformers* library, and I ran the classifier on 512 tokens at a time, which is BERT's maximum input length.

The classifier returns two results: a label (between 1 and 5, as in "4 stars"), and a score, which represents how confident the model is about the label itself. For example, if a piece of text is labelled as four stars with a score of 0.5, the model is only 50% sure that that text should be labelled as four stars.

To make comparison easier, I aggregated labels and scores with the following formula:

$$\frac{\text{amount of stars} \times \text{score}}{5}$$

Lastly, I appended the lists with all the scores to the original datasets.

⁶<https://huggingface.co/nlptown/bert-base-multilingual-uncased-sentiment>.

A code example can be seen in listing 5. De Gasperi’s median aggregated sentiment is 0.265, while Meloni’s median aggregated sentiment is 0.379. A visualization of the aggregated sentiment can be seen in figure 3. The samples are picked at random.

Listing 5. Example code for a function that calculates the sentiment score for each piece of text.

```
1 sentiment_classifier = pipeline(
2     "text-classification",
3     model="nlptown/bert-base-multilingual-uncased-sentiment",
4     device=0 # GPU
5 )
6
7 sentiment_results = sentiment_classifier(
8     texts, # list with all speeches
9     batch_size=batch_size,
10    truncation=True,
11    max_length=512 # BERT max input length
12 )
13
14 df['sentiment_aggregated'] = [
15     (int(res['label'].split()[0]) * res['score']) / 5
16     for res in sentiment_results
17 ]
18 return df
```

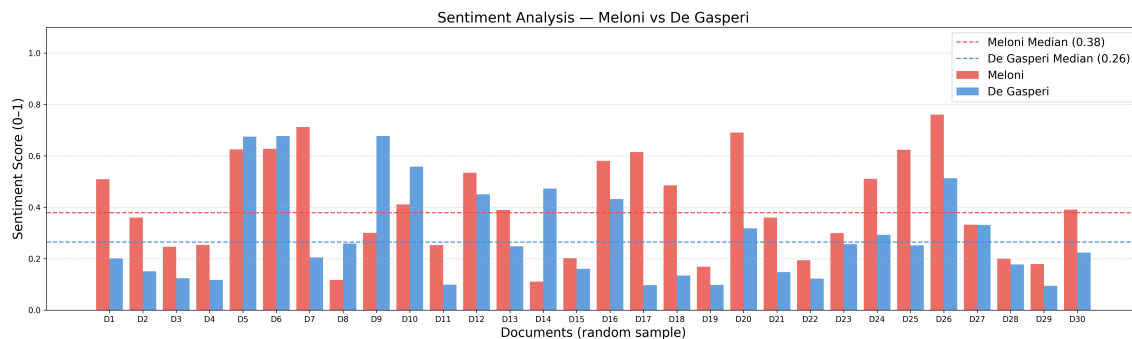


Fig. 3. Aggregated sentiment for 60 different speeches (30 for each Prime Minister), randomly sampled. The Y-axis reports the sentiment score (range between zero and one), while on the X-axis there are the relative IDs for the pairs of speeches. Each pair of speeches is represented by a two vertical bars (one red for Meloni, one blue for De Gasperi). Bar height represents their respective speech’s aggregated sentiment score. The median scores for both Prime Ministers are shown by the two horizontal dotted lines.

6.1 Limitations and Future Work

The model used to perform the sentiment analysis was designed to classify product reviews, and it struggled with long political speeches. This is evident from the sentiment scores (shown in table 5), where the model’s confidence in its classifications is around 40%.

It would be interesting to perform a sentiment analysis on the same corpus with a model trained on, for example, newspaper articles.

In future work, fine-tuning a model on labelled political speeches should produce better classification output.

Table 5. Mean and median sentiment scores of the sentiment analysis for Meloni and De Gasperi.

Metric	Meloni	De Gasperi
Mean Sentiment Score	0.4636	0.4068
Median Sentiment Score	0.4692	0.3627

7 CONCLUSION

Overall, the results are satisfactory. I managed to prototype a pipeline that retrieves YouTube videos, transcribes them in text with an encoder-decoder transformer, and saves the result in a csv file. I retrieved 72 speeches of Giorgia Meloni, for a total of 120, 116 tokens.

I compared them with the De Gasperi corpus, after extracting 473 public speeches for a total of 609, 142 tokens. I calculated the MTLT for both corpora, showing that De Gasperi’s speeches are, on average, 30% more diversified in their lexicon.

Then, I compared the two Prime Ministers’ 1000 most-used words: they have 487 words in common, with a Jaccard similarity index of 32%. The highest-frequency words in common show transversal themes, such as freedom, peace and nation, while the highest-frequency words not in common reflect the PM’s different political and societal landscapes. De Gasperi often talked about communism and socialism, while Meloni focuses on Ukraine, Trump and the Euro.

Lastly, I performed a sentiment analysis on each speech using a BERT classifier. The results underscore the importance of domain-appropriate training data: the model was trained on product reviews, not on political rhetoric, which translated to an average confidence of roughly 40% per classification.

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